

COI - AT1 - Short Answer
Test

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Course: Computer Vision

- 1) Computer Vision uses image processing, machine learning and AI Algorithms to analyze images, ~~videos~~, identify objects and make decisions based on data.
- a) • Image Acquisition and enhancement
• Object detection and recognition
• Scene Understanding
- b) • Face recognition
• Self-driving cars.
- 2) Features include image acquisition, preprocessing, feature extraction, object detection, segmentation, pattern recognition, and image classification.
- a) The process of identifying important characteristics from an image is called "feature Extraction".
- b) It enables systems to identify and classify objects accurately, improving automation and decision making.
- 3) a) Low-level image processing enhances image quality through operations like noise removal,

filtering and Contrast enhancement.

b) Mid level image processing:

- Performs Segmentation and feature Extraction
- Output: Image features

High-level image processing:-

- Performs Object recognition & Scene understanding
- Output: Meaningful interpretations or decisions.

4) A digital image is represented as a matrix of pixels, where each pixel stores intensity or color values

a) A pixel is the smallest unit of a digital image representing a single color or intensity value.

b) Higher resolution provides more detail and better image quality, while lower resolution reduces clarity.

5) A Camera sensor captures light from a scene and converts it into digital pixel values.

a) It converts incoming light into electrical signals

that are digitized into an image

- b).
- Digital Camera
 - Scanner

6. Sampling Converts the image into pixels, while quantization assigns intensity values to each pixel.

a) Sampling is the process of dividing a continuous image into discrete pixels.

b) Quantization converts continuous intensity values into discrete digital values for storage and processing.

7) It explains how light from a scene passes through a lens and forms an image on the sensor

a) Perspective projection maps a 3D scene onto a 2D image while preserving depth perception.

b) Illumination provides light that determines the brightness, contrast and visibility of objects.

8)

a) The technique of enhancing or transforming images using algorithms is called image processing.

b)

Image Processing	Computer Vision
* Enhances images * Output is an improved image	* understands image content * Output is information or decision.

9)
a) Detecting diseases such as cancer from x-rays, CT scans, or MRI images.

b) Detecting lanes, pedestrians, traffic signs

10) They provide the basic representation and quality of image required for accurate analysis.

a) Because every digital image is made up of pixels that store color and intensity information.

b) Higher sampling and resolution provide more detail, improving feature extraction.